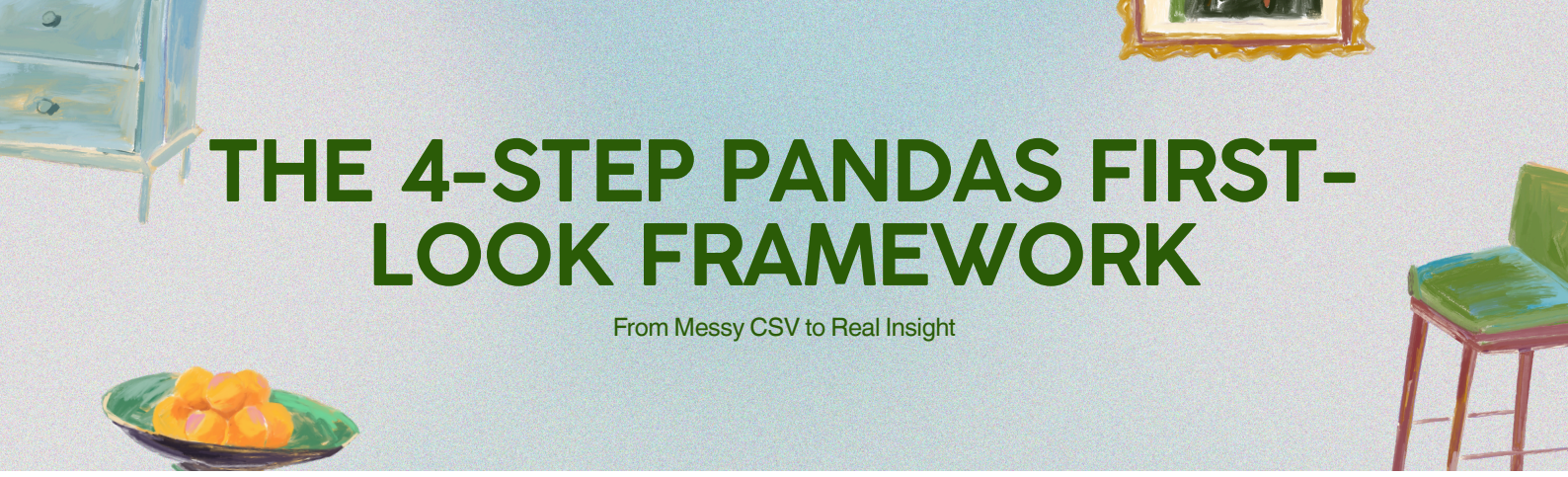




THE 4-STEP PANDAS FIRST-LOOK FRAMEWORK

From Messy CSV to Real Insight

Step 1 – Orient: Your First 60 Seconds with the Dataset

- What dataset will you work through? (file name or description):
- Run `df.info()`. How many rows and columns?
- Which columns have null values? List them:
- What dtypes look wrong (dates stored as object, numbers as string)?
- What's your first instinct about what this data is trying to tell you?

Step 2 – Select: Define the Question Before You Filter

- What is the ONE question you want to answer with this dataset?
- Which columns are essential to answering that question? List them:
- Write the `df.loc[]` or boolean filter that isolates the rows you care about:
- What rows should be excluded (test data, outliers, nulls)? Why?
- After filtering: how many rows remain? Does that feel right?

Step 3 — Clean: Audit the Mess Before You Trust the Numbers

- List every column that needs type conversion and what it should become:
- How are missing values encoded? (NaN, 'N/A', 'None', '-', blank?) List all variants found:
- Are there duplicate rows? Run `df.duplicated()` — what's the count?
- Which string columns have inconsistent casing or whitespace? (e.g. 'New York ' vs 'new york')
- Write the single cleaning step that would most improve analysis quality:

Step 4 — Aggregate: Find the Summary That Actually Tells You Something

- What column will you ? (region, category, date, customer segment?)
- What aggregation makes sense — sum, mean, count, median? Why?
- Run the `.agg()`. Which group has the highest value? Which has the lowest?
- Is any single group driving a disproportionate share (>30–40%) of the total?
- What follow-up question does this aggregation raise that the raw data couldn't?

Reflection: What the Data Actually Said

- What was the most surprising thing you found during cleaning (not analysis)?
- Did your original hypothesis hold after — or did the summary contradict it?
- What would have gone wrong if you skipped cleaning and went straight to aggregation?
- What's the one insight you'd present to a non-technical stakeholder from this exercise?
- What's the next Pandas operation you need to learn to go deeper on this dataset?